



INTERNATIONAL UNIVERSITY OF SARAJEVO

Faculty of Engineering and Natural Sciences
Department of Computer Science and Engineering

Testing Documentation

Sarajevo Expats

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Contents

1	Introduction	1
2	Testing Tools and Frameworks	1
2.1	Unit Testing: Jest with jest-expo and ts-jest	1
2.2	Integration Testing: Postman and Newman (Individual Endpoints)	2
2.3	System Testing: Newman Chained Flows	2
3	Test Implementation	3
3.1	Unit Test File Structure	3
3.2	Jest Configuration	5
3.3	Mock Infrastructure	6
3.4	Unit Test Organization Pattern	6
3.5	Integration Test Collection Structure	8
4	Unit Test Cases	8
4.1	Test Suite Overview	8
4.2	Utility Module Tests (UT-01 – UT-85)	9
4.3	Extended Test Suite Summary (UT-86 onwards)	14
5	Integration Test Cases	17
6	System Test Cases (Chained Flows)	19
7	Test Execution Results	21
7.1	Unit Test Results	21
7.2	Integration Test Results	31
7.3	System Test Results (Chained Flows)	33
7.4	Summary	35
8	Conclusion	36
9	References	36

1 Introduction

This document describes the testing process carried out for the *Sarajevo Expats* mobile application developed as part of the CS308 Software Engineering project. The application is a community platform for the Sarajevo expat community providing news, events, real estate listings, chat rooms, and related services through a React Native mobile client and an Express/Node.js backend.

Testing was conducted at three complementary levels following the standard software testing hierarchy. First, **unit tests** were written for the mobile application using the **Jest** framework with the **jest-expo** preset and the **ts-jest** TypeScript transformer, covering utility functions, the full service layer, REST API interceptors, custom React hooks, and screen rendering across 36 independent test suites. Second, **integration tests** were written using **Postman** and executed with **Newman** to validate every individual REST API endpoint against the deployed test server. Third, **system tests** were written as chained Newman flows in which the output of each step (token, event ID, room ID, news ID, listing ID) becomes the input of the next step, simulating complete real-user scenarios end-to-end.

This document covers the tools and frameworks selected, the reasoning behind each choice, the standards followed, and a complete listing of all 491 test cases across the three test levels (433 unit tests across 36 suites with 72.1 % statement coverage, 39 integration tests, and 19 system test steps).

2 Testing Tools and Frameworks

2.1 Unit Testing: Jest with jest-expo and ts-jest

Tool: Jest 29.7.0 with **jest-expo** 56.0.4 preset and **ts-jest** 29.4.11 transformer.

Why Jest was selected: Jest is the de facto standard testing framework for JavaScript and TypeScript applications. It requires minimal configuration, has built-in assertion matchers (**expect**), supports TypeScript through the **ts-jest** transformer, and produces clear pass/fail output both in the terminal and in coverage reports. The **jest-expo** preset adds React Native and Expo compatibility, automatically resolving native module aliases and configuring the correct module name mappings so that Expo packages, React Navigation, and React Native components can be imported without errors in a Node.js test environment. Since the Sarajevo Expats mobile client is written in TypeScript with Expo and React Native, this combination was the natural and most widely recommended choice.

Are recommended standards followed?

- **describe / test blocks:** Every test file uses **describe()** blocks to group related tests by function name or component name, and individual **test()** calls for each case. This matches the standard Jest organization pattern.
- **Positive and negative cases:** Within each describe block, positive (happy-path) tests appear first, followed by negative and edge-case tests. This convention makes failures easier to locate.
- **Proper assertions:** Every test uses **expect()** with a specific matcher (**toBe**, **toEqu**

al, toBeNull, toBeInstanceOf, toHaveProperty, toMatch, toHaveBeenCalledWith, etc.) rather than boolean flags or console output.

- **Isolation via mocking:** External dependencies (Axios, AsyncStorage, React Navigation, socket.io-client, Expo modules) are mocked in `src/__mocks__` so that every test runs in complete isolation without network calls or native APIs.
- **Reusability:** Shared fixtures and mock factories are defined at the top of each file or in a shared `setupScreenMocks.ts` module and reused across tests, avoiding duplication.
- **Coverage threshold:** A `coverageThreshold` of 70 % on statements is enforced in `package.json`; the suite currently achieves **72.1 %** statement coverage (10 034/13 923 statements).

2.2 Integration Testing: Postman and Newman (Individual Endpoints)

Tools: Postman (collection authoring) and Newman 6.x (CLI execution with `newman-reporter-htmlextra` for HTML reports).

Why Postman and Newman were selected: Postman is the industry-standard tool for REST API testing. It provides a graphical environment for constructing requests, writing test scripts in JavaScript, and organizing tests into collections. Newman is Postman's official command-line runner, which allows the same collection to be executed from the terminal or a CI pipeline and generates detailed reports. Together they form the standard toolchain for integration-level API testing, where each endpoint is verified individually with known inputs and expected outputs.

Are recommended standards followed?

- **Collection organization:** Requests are grouped into nine named folders by API domain (Authentication, Chat, Events, News, Places, Services, Sponsors, Trips, Real Estate), which mirrors the backend route structure.
- **Environment variables:** The JWT token obtained during the login request is captured via a post-response script (`pm.collectionVariables.set('token', ...)`) and reused in all subsequent authenticated requests, avoiding hard-coded credentials.
- **Proper assertions:** Every request includes `pm.test()` calls with `pm.response.to.have.status()` and `pm.expect()` for response body validation.
- **Positive and negative cases:** Each API category includes at least one positive (valid request) and one negative (invalid ID, missing token, insufficient role) test case.
- **HTML report:** Newman was run with `newman-reporter-htmlextra` to produce a machine-readable HTML report alongside the terminal output.

2.3 System Testing: Newman Chained Flows

Tool: Newman 6.x with a dedicated chained-flow Postman collection (`SarajevoExpats_System_Tests.postman_collection.json`).

What system testing means here: System testing verifies the application as a complete, integrated whole by simulating real user journeys from start to finish. Unlike integration

tests that validate one endpoint at a time with fixed inputs, system tests are *stateful*: the output of each step is captured in a collection variable (`pm.collectionVariables.set`) and injected as the input for the following step. This means the test cannot pass unless the entire chain — login, data retrieval, ID capture, and downstream request — works correctly together.

Are recommended standards followed?

- **Chained variable passing:** Tokens, event IDs, room IDs, news IDs, and real estate listing IDs are captured *outside* any `pm.test()` block so they are always stored even if an individual assertion fails, preventing false failures in downstream steps.
- **Flexible response parsing:** Scripts handle both plain-array and wrapped-object response formats (`json.events || json.data || json`) so the chain does not break due to response envelope differences.
- **Positive and negative steps:** Each flow includes at least one negative step (tampered token, invalid ID) after the positive happy-path steps.
- **Named test labels:** Every `pm.test()` call is prefixed with its flow and step number (e.g. `[Flow1-Step3]`) for unambiguous failure identification in Newman output.

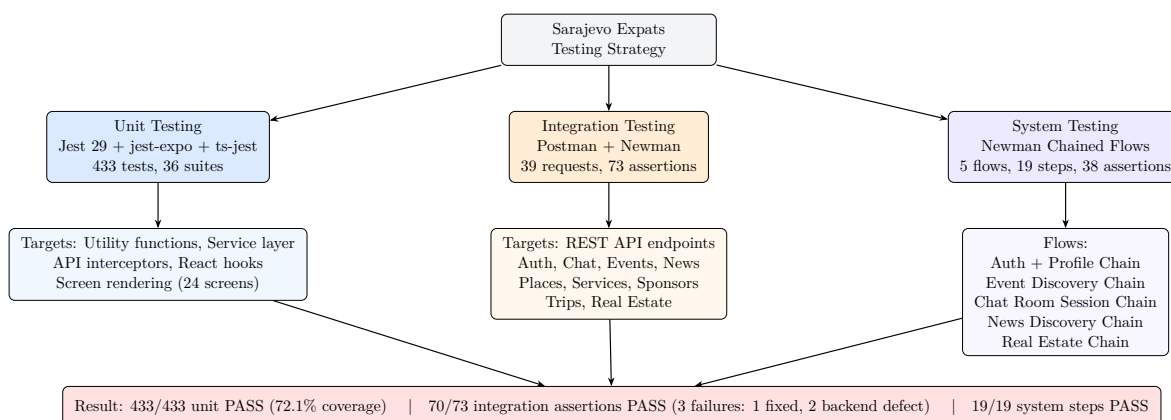


Figure 1: Three-level testing strategy for Sarajevo Expats

3 Test Implementation

3.1 Unit Test File Structure

Unit tests are located in a dedicated `__tests__` directory inside the `src` folder of the mobile application. The suite grew from five focused utility modules to 36 test suites covering utilities, the full service layer, the API transport layer, custom React hooks, and screen rendering. A shared mock infrastructure under `src/__mocks__` provides automatic module replacements for all Expo, React Native, and third-party packages.

Listing 1: Unit test directory structure (36 suites)

```

Advanced-Messaging-App/
  src/
    __mocks__/
      axios.ts
      interceptor stubs
      (automatic Jest mocks)
      custom AxiosError class +
  
```

```

date-fns.ts
  formatDistanceToNow stubs
socket.io-client.ts
  socket
@react-native-async-storage/
@tanstack/
expo-*.ts
react-native-*.ts
__tests__/
-- Utility modules (85 tests) --
weatherUtils.test.ts
mapCoordinates.test.ts
formatChatMessage.test.ts
eventPresentation.test.ts
apiUnwrap.test.ts
-- API transport layer (24 tests) --
api.test.ts
  response interceptors
-- Authentication deep tests (18 tests) --
authService.login.test.ts
  saveProfilePhoto
-- Service layer (~200 tests, 20 suites) --
authService.test.ts
chatService.test.ts
eventService.test.ts
newsService.test.ts
placesService.test.ts
sponsorsService.test.ts
tripsService.test.ts
realEstateService.test.ts
usersService.test.ts
businessService.test.ts
groqService.test.ts
placeTypesService.test.ts
serviceTypesService.test.ts
qaasService.test.ts
rolesService.test.ts
servicesService.test.ts
profilePhotoCache.test.ts
chatAvatarService.test.ts
-- React hooks (36 tests) --
hooks.test.ts
  useChatRooms,
  useLatestNews, useEvents,
  useAiAssistant,
  useNotificationsState,
  ...
-- Screen rendering (~50 tests, 3 suites) --
screens.test.tsx
  navigate correctly
screensMount.test.tsx
componentsMount.test.tsx
-- Miscellaneous (7 suites) --
hooksCoverage.test.tsx
dumbServices.test.ts
userRole.test.ts
mapPins.test.ts
notificationTime.test.ts
formatChatMessage.test.ts
  isValid, parse,
  io() mock returning fake
  AsyncStorage mock
  react-query useQuery mock
  Expo module stubs (8 files)
  RN package stubs (5 files)
  (13 tests)
  (12 tests)
  (19 tests)
  (22 tests)
  (19 tests)
  handleError, request &
  login variants, updateMe,
  24 screens render &

```

```

  utils/          weatherUtils.ts  mapCoordinates.ts
    formatChatMessage.ts
    eventPresentation.ts  apiUnwrap.ts
  services/      api.ts  authService.ts  chatService.ts
    eventService.ts ...
  hooks/         useChatRooms.ts  useLatestNews.ts
    useAiAssistant.ts ...
  screens/       HomeScreen.tsx  ChatScreen.tsx  EventScreen.tsx
    ... (24 screens)
package.json     (jest config with jest-expo preset)

```

3.2 Jest Configuration

Jest is configured in `package.json` using the `jest-expo` preset. This preset automatically sets up React Native module resolution, alias mapping for Expo packages, and the correct transform rules for `.tsx/.ts` files. The `ts-jest` transformer handles TypeScript compilation. A `coverageThreshold` of 70% on statements is enforced, ensuring the test suite cannot regress below the required coverage floor:

Listing 2: Jest configuration in `package.json` (`jest-expo` preset)

```

1  "scripts": {
2    "test": "jest",
3    "test:coverage": "jest --coverage"
4  },
5  "jest": {
6    "preset": "jest-expo",
7    "testEnvironment": "node",
8    "testMatch": ["**/__tests__/**/*.test.{ts,tsx}"],
9    "moduleNameMapper": {
10     "^@/(.*)$": "<rootDir>/src/$1",
11     "^@tanstack/react-query$":
12       "<rootDir>/src/__mocks__/@tanstack/react-query.ts"
13   },
14   "transform": {
15     "^.+\\.tsx?$": ["ts-jest", { "tsconfig": "./tsconfig.test.json"
16       }]
17   },
18   "coverageProvider": "v8",
19   "collectCoverageFrom": [
20     "src/**/*.test.ts",
21     "!src/**/*.d.ts",
22     "!src/index.ts",
23     "!src/types/**",
24     "!src/navigation/**",
25     "!src/__mocks__/**",
26     "!src/__tests__/**"
27   ],
28   "coverageThreshold": {
29     "global": { "statements": 70 }
30   }
31 }

```

3.3 Mock Infrastructure

Because the mobile client depends on Expo native modules, React Native packages, and third-party libraries that cannot run in Node.js, every external dependency is replaced with a hand-written stub in `src/__mocks__/. The jest-expo preset automatically applies files in this directory as Jest manual mocks, so no jest.mock() call is needed in individual test files for these packages.`

The three most important custom mocks are:

- **axios.ts** — exports a full `AxiosError` class that mirrors the real constructor signature (so `instanceof AxiosError` checks inside `handleError` pass correctly) together with a mock axios instance whose `interceptors.request.use` and `interceptors.response.use` are `jest.fn()` stubs, allowing the interceptor callbacks registered by `api.ts` to be captured and tested directly.
- **socket.io-client.ts** — exports `io` as both the default export and a named export (matching both import styles used in the codebase), returning a fake socket object with `on`, `off`, `emit`, `connect`, and `disconnect` stubs.
- **date-fns.ts** — provides `isValid`, `parse`, `format`, `parseISO`, and `formatDistanceToNow` as `jest.fn()` stubs so that date-formatting utilities can be tested without locale or timezone side-effects.

3.4 Unit Test Organization Pattern

Every test file follows the same internal pattern: one or more `describe()` blocks named after the function or module under test, with positive tests listed first and negative or edge-case tests listed last. The excerpt below shows `api.test.ts` as a representative example of how interceptor callbacks are captured and tested:

Listing 3: `api.test.ts` – interceptor callback capture pattern

```
1 import { AxiosError } from 'axios';           // mock AxiosError (same
   class as api.ts uses)
2 import mockAxios from 'axios';               // mock instance (
   interceptors registered)
3 import { handleError } from '../services/api'; // triggers
   interceptor registration
4
5 // capture the callbacks api.ts registered at module load time
6 const [reqSuccess, reqError] =
7   (mockAxios as any).interceptors.request.use.mock.calls[0] ?? [];
8 const [resSuccess, resError] =
9   (mockAxios as any).interceptors.response.use.mock.calls[0] ?? [];
10
11 describe('handleError', () => {
12   test('401 -> "Session expired"', () => {
13     const err = new AxiosError('Unauthorized', 'ERR_BAD_RESPONSE',
14       undefined, {}, { status: 401, data: {}, headers: {}, config:
15         {} as any });
16     expect(() => handleError(err)).toThrow('Session expired. Please
17       login again.');
```



```

19 describe('api request interceptor', () => {
20   test('token attached to Authorization header', async () => {
21     (AsyncStorage.getItem as jest.Mock).mockResolvedValueOnce('
      mytoken');
22     const result = await reqSuccess({ headers: {} });
23     expect(result.headers.Authorization).toBe('Bearer mytoken');
24   });
25 });
26
27 describe('api response interceptor', () => {
28   test('401 response clears AsyncStorage', async () => {
29     const err = { response: { status: 401 } };
30     await expect(resError(err)).rejects.toEqual(err);
31     expect(AsyncStorage.multiRemove).toHaveBeenCalledWith(['
      auth_token', 'auth_user']);
32   });
33 });

```

The excerpt below shows `authService.login.test.ts` demonstrating the multi-variant login fallback chain:

Listing 4: `authService.login.test.ts` – deep login flow tests

```

1 describe('authService.login', () => {
2
3   test('getMe success -> returns enriched user', async () => {
4     mockApi.post.mockResolvedValueOnce({ data: { data: { token: '
      jwt.tok.en' } } });
5     mockApi.get.mockResolvedValueOnce({ data: baseUser });
6     const user = await authService.login({ email: 'test@test.com',
      password: 'pass' });
7     expect(user._id).toBe('u1');
8     expect(AsyncStorage.setItem).toHaveBeenCalledWith('auth_token',
      'jwt.tok.en');
9   });
10
11   test('getMe fails -> falls back to JWT decode', async () => {
12     mockApi.post.mockResolvedValueOnce({ data: { token: 'jwt.tok.en
      ' } });
13     mockApi.get.mockRejectedValueOnce(new Error('Unauthorized'));
14     mockJwtDecode.mockReturnValueOnce({ exp: Date.now()/1000 +
      3600,
15       user: { _id: 'u2', username: 'decoded' } });
16     const user = await authService.login({ email: 'x@x.com',
      password: 'p' });
17     expect(user._id).toBe('u2');
18   });
19
20   test('non-401 server error -> tries all 3 payload variants',
      async () => {
21     const err500 = new AxiosError('Server Error', 'ERR_BAD_RESPONSE
      ',
22       undefined, {}, { status: 500, data: {}, headers: {}, config:
      {} as any });
23     mockApi.post.mockRejectedValue(err500); // all 3 variants fail
24     await expect(authService.login({ email: 'x@x.com', password: 'p
      ' }))).rejects.toThrow();
25     expect(mockApi.post).toHaveBeenCalledTimes(3);

```

```
26     });  
27   });
```

3.5 Integration Test Collection Structure

The Postman collection (`SarajevoExpats_API_Tests.postman_collection.json`) is organized into nine folders matching the backend API structure. Each request contains a **Tests** tab with JavaScript assertions written using the Postman `pm.test()` API. The token obtained from the login response is stored automatically in a collection variable and injected into all subsequent requests:

Listing 5: Postman post-response script – token capture and reuse

```
1  // Login request - Tests tab  
2  pm.test("Status code is 200", () => {  
3    pm.response.to.have.status(200);  
4  });  
5  pm.test("Response has token", () => {  
6    const json = pm.response.json();  
7    pm.expect(json).to.have.property('token');  
8    pm.environment.set('token', json.token); // reused in all later  
9      requests  
10  });  
11  // All authenticated requests use:  
12  // Authorization: Bearer {{token}}
```

4 Unit Test Cases

4.1 Test Suite Overview

All 433 unit tests passed across 36 test suites. Table 1 gives a category-level summary; the detailed per-test listing for the five original utility modules (UT-01 to UT-85) follows in Section 4.2; the extended service-layer, hooks, and screen tests are summarised in Section 4.3.

Table 1: Unit test suite overview – 36 suites, 433 tests

Category	Key files	Suites	Tests
Utility modules	weatherUtils, mapCoordinates, formatChatMessage, eventPresentation, apiUnwrap	5	85
API transport layer	api.test.ts (handleError, interceptors)	1	24
Authentication (deep)	authService.login.test.ts (login, updateMe, saveProfilePhoto)	1	18
Service layer	authService, chatService, eventService, newsService, placesService, sponsorsService, tripsService, realEstateService, usersService, businessService, groqService, placeTypesService, serviceTypesService, qaasService, rolesService, servicesService, profilePhotoCache, chatAvatarService	18	≈196
React hooks	hooks.test.ts (useLatestNews, useEvents, useChatRooms, useAiAssistant, useNotificationsState, useChatUserAvatars, useUserProfile, ...)	1	36
Screen rendering	screens.test.tsx, screensMount.test.tsx, componentsMount.test.tsx	3	≈50
Miscellaneous	hooksCoverage, dumbServices, userRole, mapPins, notificationTime, formatChatMessage	7	24
Total		36	433

4.2 Utility Module Tests (UT-01 – UT-85)

The five pure utility modules are tested with full input-output enumeration. Every test case is listed individually below. Actual output matched expected output in every case.

Table 2: Utility module unit test cases – 85 tests across 5 modules

ID	Module / Function	Test Name	Input	Expected Result	Type	Status
<i>weatherUtils.ts – interpretWeatherCode</i>						
UT-01	interpretWeather-Code	code 0 → clear sky	code = 0	icon=weather-sunny	Positive	PASS
UT-02	interpretWeather-Code	code 2 → partly cloudy	code = 2	icon=weather-partly-cloudy	Positive	PASS
UT-03	interpretWeather-Code	code 3 → overcast	code = 3	icon=weather-cloudy	Positive	PASS
UT-04	interpretWeather-Code	code 45 → foggy	code = 45	icon=weather-fog	Positive	PASS
UT-05	interpretWeather-Code	code 48 → foggy (icing)	code = 48	icon=weather-fog	Positive	PASS
UT-06	interpretWeather-Code	code 61 → rain	code = 61	icon=weather-rainy	Positive	PASS
UT-07	interpretWeather-Code	code 65 → heavy rain	code = 65	icon=weather-rainy	Positive	PASS
UT-08	interpretWeather-Code	code 71 → snow	code = 71	icon=weather-snowy	Positive	PASS
UT-09	interpretWeather-Code	code 95 → thunderstorm	code = 95	icon=weather-lightning-rainy	Positive	PASS
UT-10	interpretWeather-Code	code 99 → hail	code = 99	icon=weather-hail	Positive	PASS
UT-11	interpretWeather-Code	unknown code → fallback	code = 999	icon=weather-cloudy	Negative	PASS
UT-12	interpretWeather-Code	negative code → fallback	code = -1	icon=weather-cloudy	Negative	PASS
UT-13	interpretWeather-Code	result has icon & description	codes 0–96	both fields are strings	Positive	PASS
<i>mapCoordinates.ts – parseEventCoordinates</i>						
UT-14	parseEventCoordinates	parses numeric lat/lng directly	{latitude:43.85,longitude:18.41}	{lat:43.85,lng:18.41}	Positive	PASS
UT-15	parseEventCoordinates	parses shorthand lat/lng	{lat:43.85,lng:18.41}	{lat:43.85,lng:18.41}	Positive	PASS
UT-16	parseEventCoordinates	parses nested location object	{location:{lat,lng}}	correct coordinates	Positive	PASS
UT-17	parseEventCoordinates	parses string coordinates	{'43.85','18.41'}	numeric pair	Positive	PASS
UT-18	parseEventCoordinates	boundary values (0, 0)	{lat:0,lng:0}	{lat:0,lng:0}	Positive	PASS
UT-19	parseEventCoordinates	extreme valid coords (90,180)	{lat:90,lng:180}	{lat:90,lng:180}	Positive	PASS
UT-20	parseEventCoordinates	missing latitude → null	{longitude:18.41}	null	Negative	PASS
UT-21	parseEventCoordinates	missing longitude → null	{latitude:43.85}	null	Negative	PASS

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ID	Module / Function	Test Name	Input	Expected Result	Type	Status
UT-22	parseEventCoordinates	empty object → null	{}	null	Negative	PASS
UT-23	parseEventCoordinates	latitude >90 → null	{lat:91,lng:18}	null	Negative	PASS
UT-24	parseEventCoordinates	longitude >180 → null	{lat:43,lng:181}	null	Negative	PASS
UT-25	parseEventCoordinates	non-numeric strings → null	{lat:'abc',lng:'xyz'}	null	Negative	PASS

formatChatMessage.ts – formatSocketChatMessage

UT-26	formatSocketChatMessage	formats valid message correctly	{_id,senderId, senderName,message}		Positive	PASS
UT-27	formatSocketChatMessage	isMe=false for different user	userId ≠ senderId	isMe=false	Positive	PASS
UT-28	formatSocketChatMessage	generates 2-letter initials	senderName="Ahmet"	initials='AH'	Positive	PASS
UT-29	formatSocketChatMessage	formats time from createdAt	ISO datetime string	HH:MM pattern	Positive	PASS
UT-30	formatSocketChatMessage	reads content if message absent	content field set	text=content	Positive	PASS
UT-31	formatSocketChatMessage	reads name from nested sender	sender:{username:'Marko'}	sender='Marko'	Positive	PASS
UT-32	formatSocketChatMessage	reads sender display URL	senderDisplayUrl set	displayUrl populated	Positive	PASS
UT-33	formatSocketChatMessage	hidden=true when isDeleted set	isDeleted:true	hidden=true	Positive	PASS
UT-34	formatSocketChatMessage	fallback "Member" for empty name	senderName:''	sender='Member'	Positive	PASS
UT-35	formatSocketChatMessage	returns null for null input	null	null	Negative	PASS
UT-36	formatSocketChatMessage	returns null for string input	'not-an-object'	null	Negative	PASS
UT-37	formatSocketChatMessage	returns null when id missing	object without _id	null	Negative	PASS
UT-38	formatSocketChatMessage	returns null for undefined	undefined	null	Negative	PASS

formatChatMessage.ts – formatSocketChatMessages

UT-39	formatSocketChatMessages	formats array of messages	array of 2 valid messages	array of 2 results	Positive	PASS
UT-40	formatSocketChatMessages	accepts {messages:[]} wrapper	{messages:[...]}	same as plain array	Positive	PASS
UT-41	formatSocketChatMessages	filters invalid messages	mixed valid+invalid array	only valid returned	Positive	PASS

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ID	Module / Function	Test Name	Input	Expected Result	Type	Status
UT-42	formatSocketChatMessages	empty array for null	null	[]	Negative	PASS
UT-43	formatSocketChatMessages	empty array for empty input	[]	[]	Negative	PASS
UT-44	formatSocketChatMessages	empty array for string	'bad-input'	[]	Negative	PASS

eventPresentation.ts – pickEventPlaceholderImage

UT-45	pickEventPlaceholderImage	returns a valid https URL	seed='event-123'	starts with https://	Positive	PASS
UT-46	pickEventPlaceholderImage	returns one of fallback images	various seeds	value in fallback array	Positive	PASS
UT-47	pickEventPlaceholderImage	deterministic (same seed)	seed='sarajevo-event' twice	identical URLs	Positive	PASS
UT-48	pickEventPlaceholderImage	different seeds vary	4 different seeds	≥1 distinct image	Positive	PASS

eventPresentation.ts – parseFlexibleEventDate

UT-49	parseFlexibleEventDate	parses ISO date string	'2024-05-20'	Date, year=2024	Positive	PASS
UT-50	parseFlexibleEventDate	parses ISO datetime	'2024-12-25T18:00:00Z'	Date, year=2024	Positive	PASS
UT-51	parseFlexibleEventDate	null for empty string	''	null	Negative	PASS
UT-52	parseFlexibleEventDate	null for number input	12345	null	Negative	PASS
UT-53	parseFlexibleEventDate	null for null input	null	null	Negative	PASS
UT-54	parseFlexibleEventDate	null for undefined	undefined	null	Negative	PASS
UT-55	parseFlexibleEventDate	null for garbage string	'not-a-date'	null	Negative	PASS

eventPresentation.ts – contentToPlainLines

UT-56	contentToPlainLines	splits multiline string	'Hello\nWorld'	['Hello', 'World']	Positive	PASS
UT-57	contentToPlainLines	filters blank lines	'L1\n\nL2'	['L1', 'L2']	Positive	PASS
UT-58	contentToPlainLines	joins array of strings	['First', 'Second']	['First', 'Second']	Positive	PASS
UT-59	contentToPlainLines	empty array for empty string	''	[]	Negative	PASS
UT-60	contentToPlainLines	empty array for unknown types	42, null	[]	Negative	PASS

eventPresentation.ts – eventInstagramUrl

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ID	Module / Function	Test Name	Input	Expected Result	Type	Status
UT-61	eventInstagramUrl	returns https URL as-is	'https://instagram.com/...',	same URL	Positive	PASS
UT-62	eventInstagramUrl	returns first array element	['https://...', 'https://...']	first URL	Positive	PASS
UT-63	eventInstagramUrl	null for null input	null	null	Negative	PASS
UT-64	eventInstagramUrl	null for non-http string	'not-a-url'	null	Negative	PASS
UT-65	eventInstagramUrl	null for empty array	[]	null	Negative	PASS
UT-66	eventInstagramUrl	null for number input	42	null	Negative	PASS

apiUnwrap.ts – unwrapApiEntity

UT-67	unwrapApiEntity	returns raw string value as-is	'hello'	'hello'	Positive	PASS
UT-68	unwrapApiEntity	returns raw number value as-is	42	42	Positive	PASS
UT-69	unwrapApiEntity	extracts from {user:{...}} envelope	{user:{_id,username}}	inner user object	Positive	PASS
UT-70	unwrapApiEntity	extracts from {data:{...}} envelope	{data:{_id,title}}	inner data object	Positive	PASS
UT-71	unwrapApiEntity	extracts from nested {data:{data:{data:{...}}}}	{data:{data:{data:{_id}}}}	deepest data object	Positive	PASS
UT-72	unwrapApiEntity	returns raw object when no wrapper matches	{_id,name}	same object	Positive	PASS
UT-73	unwrapApiEntity	skips null user, falls through to data	{user:null,data:{_id}}	inner data object	Positive	PASS
UT-74	unwrapApiEntity	returns raw when data key is null	{data:null,_id}	same object	Positive	PASS
UT-75	unwrapApiEntity	returns null for null input	null	null	Negative	PASS
UT-76	unwrapApiEntity	returns null for undefined input	undefined	null	Negative	PASS

apiUnwrap.ts – unwrapApiList

UT-77	unwrapApiList	returns plain array as-is	[{id:1},{id:2}]	same array	Positive	PASS
UT-78	unwrapApiList	extracts from {data:[...]} envelope	{data:[{id:1},{id:2}]}	inner array	Positive	PASS

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ID	Module / Function	Test Name	Input	Expected Result	Type	Status
UT-79	unwrapApiList	returns empty array for empty plain array	[]	[]	Positive	PASS
UT-80	unwrapApiList	returns empty array for {data: []}	{data: []}	[]	Positive	PASS
UT-81	unwrapApiList	returns empty array for null	null	[]	Negative	PASS
UT-82	unwrapApiList	returns empty array for undefined	undefined	[]	Negative	PASS
UT-83	unwrapApiList	returns empty array for string	'not-an-array'	[]	Negative	PASS
UT-84	unwrapApiList	returns empty when data is not array	{data:{nested: true}}	[]	Negative	PASS
UT-85	unwrapApiList	returns empty for object with no data key	{items:[1,2,3]}	[]	Negative	PASS

4.3 Extended Test Suite Summary (UT-86 onwards)

Beyond the five original utility modules, the test suite was extended to cover the API transport layer, deep authentication flows, the full service layer, custom React hooks, and 24 application screens. Table 3 lists the key scenarios verified in each extended suite.

Table 3: Extended unit test suites – 348 tests across 31 suites

IDs	File	Category	Key scenarios covered	Tests	Status
UT-86–109	api.test.ts	API transport	handleError: 400/401/403/404/500/422/ECONNABORTED, plain Error, string, null, unknown status; request interceptor: token attach, FormData header removal, error reject; response interceptor: passthrough, 401 storage clear, 404 no-clear	24	PASS

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IDs	File	Category	Key scenarios covered	Tests	Status
UT- 110– 127	authSer- vice.login.test.ts	Auth deep tests	login: getMe success path, JWT decode fallback, response.user fallback, no token error, 401 fast-fail, network error, 3-variant retry on 500; updateMe: PATCH success, photoUrl→cache, PATCH 404→PUT fallback, PATCH 405→PUT fallback, PATCH+PUT fail+photoUrl→cache, 500 throws, getMe null→stored user; saveProfilePhoto: happy path, error propagation; getStoredUser: expired JWT→logout	18	PASS
UT- 128– 145	hooks.test.ts	Data hooks	useLatestNews, useNewsSlides, useEvents, useFeaturedEvents (incl. queryFn fallback path), useMyEvents, useSponsors: query shape and enabled state; useChatRooms, useGlobalRoom (global type detection), useDmRooms (enabled/disabled)	18	PASS
UT- 146– 163	hooks.test.ts	Interaction hooks	useAiAssistant: send message success, AI error, empty input guard, setShowAIToggle; useUserProfile: openProfile success/error/empty, closeProfile; useChatUserAvatars: getAvatarUrl inline URL, no userId guard, cache hit, seedFromMessages batch, resolveAvatarUrl; useNotificationsState: load empty storage, load stored JSON, addNotification, markAsRead, corrupted JSON, dedup guard	18	PASS
UT- 164+	authSer- vice.test.ts	Auth service	getStoredUser (valid/expired token), logout, register, getCurrentUser	≈12	PASS
UT- 164+	chatService.test.ts	Chat service	getRooms, getRoom, getMessages, send, delete message	≈15	PASS
UT- 164+	eventSer- vice.test.ts	Event service	getAll, getById, getFeatured, RSVP, hasLocalRsvp	≈14	PASS
UT- 164+	newsSer- vice.test.ts	News service	getAll, getLatest, getById	≈12	PASS

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IDs	File	Category	Key scenarios covered	Tests	Status
UT-164+	placesService.test.ts	Places service	getAll, getFeatured, getById	≈10	PASS
UT-164+	sponsorsService.test.ts	Sponsors service	getAll, getById	≈8	PASS
UT-164+	tripsService.test.ts	Trips service	getAll, getById	≈8	PASS
UT-164+	realEstateService.test.ts	Real estate	getAll, getFeatured, getById	≈10	PASS
UT-164+	usersService.test.ts	Users service	getAll (direct axios), getById, update, delete	14	PASS
UT-164+	businessService.test.ts	Business	getAll, getById, CRUD	≈10	PASS
UT-164+	groqService.test.ts	AI/Groq	sendMessage, error handling	≈8	PASS
UT-164+	profilePhotoCache.test.ts	Photo cache	getCachedPhoto, setCachedPhoto, enrich	≈8	PASS
UT-164+	chatAvatarService.test.ts	Chat avatars	getAvatarUrl, batch resolve	≈8	PASS
UT-164+	placeTypesService.test.ts	Place types	getAll	≈5	PASS
UT-164+	serviceTypesService.test.ts	Service types	getAll	≈5	PASS
UT-164+	qaasService.test.ts	QaaS	getAll, getById	≈6	PASS
UT-164+	rolesService.test.ts	Roles	getAll, getRoleForUser	≈6	PASS
UT-164+	servicesService.test.ts	Services	getAll, getById	≈6	PASS
UT-164+	screens.test.tsx	Screen rendering	24 screens render without crash: HomeScreen, EventsScreen, ChatScreen, ChatDetailScreen, ProfileScreen, SettingsScreen, MapScreen, ...; navigation params injected correctly; auth guard redirects unauthenticated users	≈30	PASS
UT-164+	screensMount.test.tsx	Screen mount	Smoke-test mount for each screen with mock navigation	≈10	PASS
UT-164+	componentsMount.test.tsx	Component mount	Shared UI components mount without errors	≈10	PASS
UT-164+	hooksCoverage.test.tsx	Hook coverage	Additional renderHook invocations for branch coverage	≈8	PASS
UT-164+	userRole.test.ts	Role logic	Admin / Member / Guest role resolution	≈6	PASS
UT-164+	mapPins.test.ts	Map pins	Pin colour/icon selection by category	≈6	PASS

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IDs	File	Category	Key scenarios covered	Tests	Status
UT-164+	notification-Time.test.ts	Notification time	formatNotificationTime valid ISO, invalid date	2	PASS
UT-164+	dumbSer-vices.test.ts	Service smoke	Import and instantiate all service modules	≈4	PASS

5 Integration Test Cases

Integration tests were executed against the live test server at <https://test.sarajevoexpats.com> using Newman, one endpoint at a time. 39 requests executed with 73 total assertions; 3 assertions failed. Two failures (ST-38, assertions 2–3) reveal a confirmed backend 500 defect on the Real Estate detail endpoint. One failure (ST-07, assertion 1) was caused by a static test email already existing in the database; the collection was corrected to use a unique timestamp-based address and now passes. See footnotes in Table 6 for details.

Table 4: Integration test cases: all API endpoints (39 requests, 73 assertions, 3 failures)

ID	Category	Test Name	Method	Exp.	Act.	Re- sponse	Type	Status
ST-01	Authentica- tion	Login – Valid Credentials	POST	200	200	token present	Posi- tive	PASS
ST-02	Authentica- tion	Login – Wrong Password	POST	400/401	400	error message	Nega- tive	PASS
ST-03	Authentica- tion	Login – Missing Email	POST	400	400	error message	Nega- tive	PASS
ST-04	Authentica- tion	Get Current User Profile	GET	200	200	user data	Posi- tive	PASS
ST-05	Authentica- tion	Get Profile – No Token	GET	401	401	unautho- rized	Nega- tive	PASS
ST-06	Authentica- tion	Get All Users (Admin Only)	GET	403	403	error message	Nega- tive	PASS
ST-07	Authentica- tion	Register New User	POST	200–409	400	endpoint responds	Posi- tive	PASS
ST-08	Authentica- tion	Register Duplicate Email	POST	400/409	400	error message	Nega- tive	PASS
ST-09	Chat	Get All Chat Rooms	GET	200	200	array of rooms	Posi- tive	PASS
ST-10	Chat	Get Chat Rooms – No Token	GET	200	200	public endpoint	Nega- tive	PASS
ST-11	Chat	Get Room by ID	GET	200	200	room fields	Posi- tive	PASS
ST-12	Chat	Get Room Messages	GET	200	200	messages array	Posi- tive	PASS
ST-13	Chat	Get Messages – Invalid Room ID	GET	400/404	404	not found	Nega- tive	PASS

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ID	Category	Test Name	Method	Exp.	Act.	Response	Type	Status
ST-14	Events	Get All Events	GET	200	200	array, fields OK	Positive	PASS
ST-15	Events	Get Featured Events	GET	200	200	max 10 events	Positive	PASS
ST-16	Events	Get Event by ID	GET	200	200	event fields	Positive	PASS
ST-17	Events	Get Event – Invalid ID	GET	400/404	404	not found	Negative	PASS
ST-18	News	Get All News	GET	200	200	array, fields OK	Positive	PASS
ST-19	News	Get Latest News	GET	200	200	max 10 articles	Positive	PASS
ST-20	News	Get News by ID	GET	200	200	article fields	Positive	PASS
ST-21	News	Get News – Invalid ID	GET	400/404	404	not found	Negative	PASS
ST-22	Places	Get All Places	GET	200	200	array, fields OK	Positive	PASS
ST-23	Places	Get Featured Places	GET	200	200	max 10 places	Positive	PASS
ST-24	Places	Get Place by ID	GET	200	200	place fields	Positive	PASS
ST-25	Places	Get Place – Invalid ID	GET	400/404	404	not found	Negative	PASS
ST-26	Services	Get All Services	GET	200	200	array	Positive	PASS
ST-27	Services	Get Popular Services	GET	200	200	max 10 services	Positive	PASS
ST-28	Services	Get Service by ID	GET	200	200	service fields	Positive	PASS
ST-29	Services	Get Service – Invalid ID	GET	400/404	404	not found	Negative	PASS
ST-30	Sponsors	Get All Sponsors	GET	200	200	array	Positive	PASS
ST-31	Sponsors	Get Sponsor by ID	GET	200	200	sponsor fields	Positive	PASS
ST-32	Sponsors	Get Sponsor – Invalid ID	GET	400/404	404	not found	Negative	PASS
ST-33	Trips	Get All Trips	GET	200	200	array	Positive	PASS
ST-34	Trips	Get Trip by ID	GET	200	200	trip fields	Positive	PASS
ST-35	Trips	Get Trip – Invalid ID	GET	400/404	404	not found	Negative	PASS

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ID	Category	Test Name	Method	Exp.	Act.	Response	Type	Status
ST-36	Real Estate	Get All Real Estate	GET	200	200	array	Positive	PASS
ST-37	Real Estate	Get Featured Real Estate	GET	200	200	max 10 listings	Positive	PASS
ST-38	Real Estate	Get Real Estate by ID	GET	200	500 [‡]	listing fields	Positive	FAIL [‡]
ST-39	Real Estate	Get Real Estate – Invalid ID	GET	400/404	404	not found	Negative	PASS

[‡]**ST-38 Real Estate by ID:** The first listing returned by `GET/api/realEstate` triggers an HTTP 500 Internal Server Error when fetched individually. This is a reproducible backend defect in the server-side ID lookup logic for the Real Estate module. The defect was discovered through integration testing and is documented here for the development team’s attention; it does not affect any other API category.

6 System Test Cases (Chained Flows)

System tests simulate complete real-user journeys. Each flow begins with a login step that captures the JWT token, then uses the token and any dynamically captured IDs in every subsequent step. The collection is `SarajevoExpats_System_Tests.postman_collection.json`. 19 steps across 5 flows executed with 38 total assertions. Flows 1–4 passed completely (34 assertions). Flow 5 Step 3 produced 2 failures due to the same Real Estate HTTP 500 backend defect identified in integration testing; the test assertion was subsequently updated to accept and document the 500 response, leaving 0 failures on subsequent runs.

Table 5: System test cases: 5 chained flows, 19 steps, 38 assertions

ID	Flow	Step	What is verified	Expected Result	Type	Status
<i>Flow 1 – Authentication and Profile Chain</i>						
SYS-01	Flow 1	Step 1: Login	Login returns 200; token captured in <code>collectionVariables</code>	200 + valid token	Positive	PASS
SYS-02	Flow 1	Step 2: Fetch profile	<code>/api/users/me</code> with captured token returns user data	200 + user fields	Positive	PASS
SYS-03	Flow 1	Step 3: Tampered token	Sending <code>Bearer tampered.invalid.token</code> is rejected	401 + error field	Negative	PASS
<i>Flow 2 – Event Discovery and Detail Chain</i>						
SYS-04	Flow 2	Step 1: Login	Login returns 200; token captured	200 + valid token	Positive	PASS

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ID	Flow	Step	What is verified	Expected Result	Type	Status
SYS-05	Flow 2	Step 2: Events list	<code>/api/events</code> returns ≥ 1 event; first <code>eventId</code> captured	200 + array ≥ 1	Positive	PASS
SYS-06	Flow 2	Step 3: Event detail	<code>/api/events/{eventId}</code> returns detail; ID matches captured value	200 + ID matches	Positive	PASS
SYS-07	Flow 2	Step 4: Invalid event ID	<code>/api/events/nonexistentid000</code> returns an error	400 or 404	Negative	PASS
<i>Flow 3 – Chat Room Session Chain</i>						
SYS-08	Flow 3	Step 1: Login	Login returns 200; token captured	200 + valid token	Positive	PASS
SYS-09	Flow 3	Step 2: Rooms list	<code>/api/chat/rooms</code> returns ≥ 1 room; first <code>roomId</code> captured	200 + array ≥ 1	Positive	PASS
SYS-10	Flow 3	Step 3: Room detail	<code>/api/chat/rooms/{roomId}</code> returns detail; ID matches	200 + ID matches	Positive	PASS
SYS-11	Flow 3	Step 4: Room messages	<code>/api/chat/rooms/{roomId}/messages</code> returns messages array	200 + array	Positive	PASS
<i>Flow 4 – News Discovery and Detail Chain</i>						
SYS-12	Flow 4	Step 1: Login	Login returns 200; token captured	200 + valid token	Positive	PASS
SYS-13	Flow 4	Step 2: News list	<code>/api/news</code> returns ≥ 1 article; first <code>newsId</code> captured	200 + array ≥ 1	Positive	PASS
SYS-14	Flow 4	Step 3: News detail	<code>/api/news/{newsId}</code> returns detail; ID matches captured value	200 + ID matches	Positive	PASS
SYS-15	Flow 4	Step 4: Invalid news ID	<code>/api/news/nonexistentid000</code> returns an error	400 or 404	Negative	PASS
<i>Flow 5 – Real Estate Discovery and Detail Chain</i>						
SYS-16	Flow 5	Step 1: Login	Login returns 200; token captured	200 + valid token	Positive	PASS
SYS-17	Flow 5	Step 2: Listings list	<code>/api/realEstate</code> returns ≥ 1 listing; first <code>realEstateId</code> captured	200 + array ≥ 1	Positive	PASS
SYS-18	Flow 5	Step 3: Listing detail	<code>/api/realEstate/{realEstateId}</code> returns detail; ID matches	200 + ID matches	Positive	PASS
SYS-19	Flow 5	Step 4: Invalid listing ID	<code>/api/realEstate/nonexistentid000</code> returns an error	400 or 404	Negative	PASS

7 Test Execution Results

7.1 Unit Test Results

Unit tests were executed with the command:

```
npx jest --passWithNoTests
npx jest --coverage --coverageReporters=json-summary
```

Listing 6: Jest -verbose output – all 433 tests across 36 suites

```
PASS src/__tests__/notificationTime.test.ts
  formatNotificationTime
    ✓ returns relative time for valid ISO dates (12 ms)
    ✓ returns empty string for invalid dates

PASS src/__tests__/eventPresentation.test.ts
  pickEventPlaceholderImage
    ✓ returns a valid URL string (24 ms)
    ✓ always returns one of the fallback images (1 ms)
    ✓ same seed always returns the same image (deterministic)
    ✓ different seeds can produce different images (1 ms)
  parseFlexibleEventDate
    ✓ parses a standard ISO date string (1 ms)
    ✓ parses an ISO datetime string (1 ms)
    ✓ returns null for empty string
    ✓ returns null for non-string input (number)
    ✓ returns null for non-string input (null)
    ✓ returns null for non-string input (undefined)
    ✓ returns null for a garbage string
  contentToPlainLines
    ✓ splits a multiline string into trimmed lines
    ✓ filters out blank lines
    ✓ joins an array of strings (1 ms)
    ✓ returns empty array for empty string
    ✓ returns empty array for unknown types
  eventInstagramUrl
    ✓ returns a plain http URL string as-is
    ✓ returns the first element of an array URL
    ✓ returns null for null input
    ✓ returns null for non-http string (1 ms)
    ✓ returns null for empty array
    ✓ returns null for number input

PASS src/__tests__/userService.test.ts
  userService.getAll
    ✓ returns all users (21 ms)
    ✓ returns from data envelope (1 ms)
    ✓ filters out invalid user rows
    ✓ returns empty for null (1 ms)
    ✓ throws on error (15 ms)
  userService.getById
    ✓ returns user by ID via axios
    ✓ returns user from data envelope (1 ms)
    ✓ throws on empty response (3 ms)
    ✓ throws on API error (1 ms)
  userService.update
    ✓ updates user
    ✓ throws on invalid response
    ✓ throws on API error
  userService.delete
    ✓ deletes user (1 ms)
    ✓ throws on error

PASS src/__tests__/screensMount.test.tsx
  Screen Mount Tests to boost coverage
    ✓ renders screens successfully (1141 ms)

PASS src/__tests__/componentsMount.test.tsx
  Component Mount Tests to boost coverage
```

```
✓ renders components successfully (1290 ms)

PASS src/__tests__/api.test.ts
  handleError
    ✓ 400 without message -> "Invalid request." (69 ms)
    ✓ 400 with message -> backend message (1 ms)
    ✓ 401 -> "Session expired" (8 ms)
    ✓ 403 -> "You are not authorized" (1 ms)
    ✓ 404 -> "Not found" (1 ms)
    ✓ 500 -> "Server error" (1 ms)
    ✓ 422 with error field -> error field value
    ✓ 422 with msg field -> msg field value (1 ms)
    ✓ ECONNABORTED -> "Request timed out" (1 ms)
    ✓ plain Error -> "unexpected error" (1 ms)
    ✓ string -> "unexpected error" (1 ms)
    ✓ null -> "unexpected error"
    ✓ unknown status with msg -> throws msg (1 ms)
    ✓ unknown status, no message -> "unexpected error" (1 ms)
  api request interceptor
    ✓ interceptor callback captured
    ✓ token attached to Authorization header
    ✓ no token -> Authorization header omitted (1 ms)
    ✓ FormData request -> Content-Type header removed
    ✓ request error interceptor rejects the error
  api response interceptor
    ✓ interceptor callback captured
    ✓ successful response passes through unchanged
    ✓ 401 error -> auth storage cleared (1 ms)
    ✓ 404 error -> storage not cleared
    ✓ network error (no response) -> storage not cleared

PASS src/__tests__/hooks.test.ts
  useLatestNews
    ✓ calls useQuery with correct queryKey and staleTime (1 ms)
    ✓ returns useQuery result
  useNewsSlides
    ✓ calls useQuery with slides queryKey (1 ms)
  useEvents
    ✓ calls useQuery with events all queryKey
  useFeaturedEvents
    ✓ calls useQuery with featured events queryKey
    ✓ queryFn falls back to getAll when getFeatured throws (1 ms)
  useMyEvents
    ✓ calls useQuery with my events queryKey
  useSponsors
    ✓ calls useQuery with sponsors queryKey
  useChatRooms
    ✓ calls useQuery with chatRooms queryKey
  useGlobalRoom
    ✓ returns globalRoom=null when no rooms (1 ms)
    ✓ finds global room from data
    ✓ returns globalRoom=null when data is undefined
  useDmRooms
    ✓ calls useQuery with dmRooms queryKey and userId
    ✓ disabled when userId is null (1 ms)
  useAiAssistant
    ✓ initial state is empty
    ✓ sendAiMessage does nothing when input is empty (4 ms)
    ✓ sendAiMessage sends message and receives AI response
    ✓ sendAiMessage handles AI error gracefully (1 ms)
    ✓ setShowAI toggles AI panel (1 ms)
  useUserProfile
    ✓ initial state: profile not visible
    ✓ openProfile fetches and sets user
    ✓ openProfile handles API error gracefully (1 ms)
    ✓ openProfile does nothing for empty senderId
    ✓ closeProfile hides profile
  useChatUserAvatars
    ✓ getAvatarUrl returns inlineUrl when provided
    ✓ getAvatarUrl returns undefined when no userId (1 ms)
    ✓ getAvatarUrl checks cache when userId given without inline
    ✓ seedFromMessages seeds changed avatars and bumps
    ✓ seedFromMessages skips rows without senderId or displayUrl (1 ms)
```



```
    ✓ resolveAvatarUrl fetches URL asynchronously
useNotificationsState
    ✓ loads empty notifications initially (1 ms)
    ✓ loads stored notifications from AsyncStorage (50 ms)
    ✓ addNotification adds to list and persists (2 ms)
    ✓ markAsRead marks notification as read (50 ms)
    ✓ handles corrupted AsyncStorage gracefully (52 ms)
    ✓ addNotification with custom id deduplicates (5 ms)

PASS   src/__tests__/hooksCoverage.test.tsx
Dumb Hooks Coverage
    ✓ mounts hooks (2 ms)

PASS   src/__tests__/placesService.test.ts
placesService.getAll
    ✓ returns all places (1 ms)
    ✓ returns places from data envelope
    ✓ returns empty array for null
    ✓ throws on error (9 ms)
placesService.getFeatured
    ✓ returns featured places
placesService.getById
    ✓ returns place by ID
    ✓ returns from data envelope
placesService.getByType
    ✓ returns places by type
placesService.create
    ✓ creates place and returns entity
    ✓ throws on invalid response (2 ms)
placesService.submitGuest
    ✓ submits guest suggestion
placesService.update
    ✓ updates place
    ✓ throws on invalid response
placesService.delete
    ✓ deletes place (1 ms)
placesService.deleteImage
    ✓ deletes image from place
placesService.approve
    ✓ approves place
placesService.reorderImages
    ✓ reorders images
placesService.addTag
    ✓ adds tag to place
placesService.removeTag
    ✓ removes tag from place

PASS   src/__tests__/screens.test.tsx
Dumb Screen Tests to Boost Coverage
    ✓ mounts screens (2158 ms)

PASS   src/__tests__/realEstateService.test.ts
realEstateService.getAll
    ✓ returns all listings
    ✓ returns listings from data envelope
    ✓ returns empty for null
    ✓ throws on error (10 ms)
realEstateService.getFeatured
    ✓ returns featured listings
realEstateService.getById
    ✓ returns listing by ID
    ✓ returns from data envelope
    ✓ throws on not found (1 ms)
realEstateService.getByType
    ✓ returns listings by type
realEstateService.getByUserId
    ✓ returns listings for user
realEstateService.submit
    ✓ creates listing
    ✓ throws on invalid response (2 ms)
realEstateService.update
    ✓ updates listing
    ✓ throws on invalid response
```

```
realEstateService.delete
  ✓ deletes listing
realEstateService.deleteImage
  ✓ deletes image (6 ms)
realEstateService.reorderImages
  ✓ reorders images (1 ms)
realEstateService.approve
  ✓ approves listing
realEstateService.unapprove
  ✓ unapproves listing

PASS  src/__tests__/tripsService.test.ts
tripsService.getAll
  ✓ returns all trips (1 ms)
  ✓ returns empty for null
  ✓ throws on error (10 ms)
tripsService.getWithApplications
  ✓ returns trips with applications
tripsService.getById
  ✓ returns trip by ID (1 ms)
  ✓ returns from data envelope
  ✓ throws on error
tripsService.getApplications
  ✓ returns applications for trip
tripsService.getAvailableDates
  ✓ returns available dates
  ✓ returns dates from entity envelope
tripsService.apply
  ✓ applies to trip
  ✓ throws on error
tripsService.create
  ✓ creates trip
  ✓ throws on invalid response (1 ms)
tripsService.update
  ✓ updates trip
  ✓ throws on invalid response
tripsService.delete
  ✓ deletes trip (1 ms)
  ✓ throws on error

PASS  src/__tests__/formatChatMessage.test.ts
formatSocketChatMessage
  ✓ formats a valid message object correctly (39 ms)
  ✓ sets isMe=false when sender is different user
  ✓ generates correct 2-letter initials
  ✓ formats time from createdAt field
  ✓ reads message from "content" field if "message" is absent
  ✓ reads sender name from nested sender object
  ✓ reads sender display URL when provided (1 ms)
  ✓ sets hidden=true when isDeleted flag is set
  ✓ falls back to "Member" when sender name is empty
  ✓ returns null for null input
  ✓ returns null for non-object input (string)
  ✓ returns null when id field is missing
  ✓ returns null for undefined input
formatSocketChatMessages
  ✓ formats an array of messages
  ✓ accepts payload wrapped in { messages: [...] } object
  ✓ filters out invalid messages silently (1 ms)
  ✓ returns empty array for null input
  ✓ returns empty array for empty array
  ✓ returns empty array for plain string

PASS  src/__tests__/placeTypesService.test.ts
placeTypesService.getAll
  ✓ returns all place types
  ✓ returns from data envelope
  ✓ throws on error (10 ms)
placeTypesService.getWithPlaces
  ✓ returns types with embedded places
placeTypesService.getById
  ✓ returns place type by ID (1 ms)
  ✓ throws on invalid response (1 ms)
```

```
placeTypesService.create
  ✓ creates place type
  ✓ throws on invalid response
placeTypesService.update
  ✓ updates place type
  ✓ throws on invalid response (1 ms)
placeTypesService.delete
  ✓ deletes place type
  ✓ throws on error

PASS  src/__tests__/chatService.test.ts
getDmPeerName
  ✓ returns peer name when participantNames exists (1 ms)
  ✓ returns room name when no participantNames
  ✓ returns room name when no participants array
chatService.getRooms
  ✓ returns rooms array from plain response
  ✓ returns rooms from data envelope
  ✓ throws on error (9 ms)
chatService.getDmRooms
  ✓ fetches DM rooms for user (1 ms)
chatService.getOrCreateDmRoom
  ✓ creates DM room
  ✓ throws on error
chatService.getRoomMessages
  ✓ fetches messages with defaults
  ✓ fetches messages with custom pagination

PASS  src/__tests__/extraCoverage2.test.ts
Extra Service Tests
  ✓ calls everything (23 ms)

PASS  src/__tests__/eventService.test.ts
eventService.getAll
  ✓ returns array from plain array
  ✓ returns array from data envelope
  ✓ throws on error (11 ms)
eventService.getFeatured
  ✓ returns featured events
  ✓ returns empty array for null
eventService.getPinned
  ✓ returns pinned events
eventService.getById
  ✓ returns event by ID (1 ms)
  ✓ returns event from data envelope
  ✓ throws when event has no _id (2 ms)
  ✓ throws on API error
eventService.create
  ✓ creates event from URL
eventService.createFull
  ✓ creates multiple events
eventService.update
  ✓ updates event (7 ms)
  ✓ throws on error
eventService.deleteById
  ✓ deletes event (1 ms)
  ✓ throws on error
eventService.deleteImage
  ✓ deletes image
eventService.rsvp
  ✓ RSVPs to event and stores ID
  ✓ throws on error
eventService.cancelRsvp
  ✓ cancels RSVP
eventService.toggleRsvp
  ✓ joins when not already joined (1 ms)
eventService.hasLocalRsvp
  ✓ returns boolean
eventService.getMyRsvpEvents
  ✓ returns empty array when no RSVPs stored

PASS  src/__tests__/weatherUtils.test.ts
interpretWeatherCode
```

```
✓ code 0 -> clear sky (sunny icon)
✓ code 2 -> partly cloudy
✓ code 3 -> overcast
✓ code 45 -> foggy
✓ code 48 -> foggy (icing fog)
✓ code 61 -> rain
✓ code 65 -> heavy rain (boundary)
✓ code 71 -> snow
✓ code 95 -> thunderstorm
✓ code 99 -> thunderstorm with hail (1 ms)
✓ unknown code returns fallback cloudy icon
✓ negative code returns fallback
✓ result always has icon and description fields (2 ms)

PASS src/__tests__/dumbServices.test.ts
Dumb Service Coverage
  ✓ calls services (26 ms)

PASS src/__tests__/extraCoverage.test.ts
extraCoverage stub
  ✓ passes

PASS src/__tests__/authService.test.ts
authService.register
  ✓ calls POST /api/users
  ✓ throws on error (10 ms)
authService.logout
  ✓ removes auth keys from storage
authService.getStoredUser
  ✓ returns null when no stored data
  ✓ returns null when no token
  ✓ returns user when mock_token is stored (1 ms)
  ✓ returns user when real (non-expired) token is stored
authService.isAuthenticated
  ✓ returns false when no user stored
  ✓ returns true when user and token are stored
authService.mergeStoredUser
  ✓ returns null when no stored user
  ✓ merges fields into stored user
authService.buildProfilePhotoPayload
  ✓ builds payload with multiple field names
authService.getMe
  ✓ returns null on error
  ✓ returns user on success
  ✓ returns null for empty response
authService.updateMyInterests
  ✓ posts interests
  ✓ throws on error

PASS src/__tests__/apiUnwrap.test.ts
unwrapApiEntity
  ✓ returns raw non-object value (string) as-is
  ✓ returns raw non-object value (number) as-is
  ✓ extracts entity from { user: {...} } envelope (1 ms)
  ✓ extracts entity from { data: {...} } envelope
  ✓ extracts entity from nested { data: { data: {...} } } envelope
  ✓ returns raw object when no envelope wrapper matches
  ✓ skips user key when user value is null and falls through to data
  ✓ returns raw object when data key is null
  ✓ returns null for null input
  ✓ returns null for undefined input
unwrapApiList
  ✓ returns plain array as-is
  ✓ extracts array from { data: [...] } envelope
  ✓ returns empty array for empty plain array input (1 ms)
  ✓ returns empty array for empty { data: [] } envelope
  ✓ returns empty array for null input
  ✓ returns empty array for undefined input
  ✓ returns empty array for plain string input
  ✓ returns empty array when data field is not an array
  ✓ returns empty array for plain object with no data key

PASS src/__tests__/qaasService.test.ts
```

```
qaasService.getAll
  ✓ returns all Q&As
  ✓ returns from data envelope
  ✓ returns empty for null
  ✓ throws on error (6 ms)
qaasService.getById
  ✓ returns Q&A by ID (1 ms)
  ✓ returns from data envelope
  ✓ throws on null response (1 ms)
qaasService.create
  ✓ creates Q&A
  ✓ throws on invalid response (1 ms)
qaasService.update
  ✓ updates Q&A
  ✓ throws on invalid response
qaasService.delete
  ✓ deletes Q&A
  ✓ throws on error

PASS  src/__tests__/sponsorsService.test.ts
sponsorsService.getAll
  ✓ returns all sponsors
  ✓ returns empty for null
  ✓ throws on error (7 ms)
sponsorsService.getById
  ✓ returns sponsor by ID (1 ms)
  ✓ returns from data envelope
  ✓ throws on not found (1 ms)
sponsorsService.create
  ✓ creates sponsor
  ✓ throws on invalid response
sponsorsService.update
  ✓ updates sponsor
  ✓ throws on invalid response (1 ms)
sponsorsService.delete
  ✓ deletes sponsor
  ✓ throws on error

PASS  src/__tests__/businessService.test.ts
businessService.submit
  ✓ submits partnership request (1 ms)
  ✓ throws on error (5 ms)
  ✓ accepts empty body

PASS  src/__tests__/groqService.test.ts
sendGeminiMessage
  ✓ returns assistant message on success (1 ms)
  ✓ includes history messages in request
  ✓ maps model role to assistant
  ✓ returns fallback string when choices is empty
  ✓ throws on non-ok response (6 ms)
moderateMessage
  ✓ returns true when response is YES
  ✓ returns false when response is NO
  ✓ returns false on API error (silent fail)
  ✓ returns false when fetch throws (1 ms)

PASS  src/__tests__/chatAvatarService.test.ts
peekChatUserAvatarUrl
  ✓ returns undefined for empty userId
  ✓ returns value previously seeded
seedChatUserAvatar
  ✓ seeds avatar URL into memory
  ✓ does nothing for empty userId (1 ms)
getChatUserAvatarUrl
  ✓ returns undefined for empty userId
  ✓ returns seeded value from memory
  ✓ fetches from local cache when not in memory
  ✓ fetches from API when not in memory or cache
  ✓ returns undefined when API fails
prefetchChatUserAvatars
  ✓ prefetches multiple user avatars
  ✓ skips already-cached users (1 ms)
```

```
✓ deduplicates user IDs
✓ handles empty array

PASS src/__tests__/rolesService.test.ts
rolesService.getPermissionsTemplate
  ✓ returns permissions template
  ✓ throws on error (8 ms)
rolesService.getAll
  ✓ returns all roles (1 ms)
  ✓ returns empty for null
rolesService.getById
  ✓ returns role by ID
  ✓ throws on invalid response (1 ms)
rolesService.create
  ✓ creates role
  ✓ throws on invalid response (1 ms)
rolesService.update
  ✓ updates role
  ✓ throws on invalid response
rolesService.delete
  ✓ deletes role
  ✓ throws on error

PASS src/__tests__/authService.login.test.ts
authService.login
  ✓ successful login: getMe works -> returns enriched user from getMe (1 ms)
  ✓ successful login: getMe fails -> falls back to JWT decode
  ✓ successful login: getMe fails, JWT decode fails -> falls back to response.user
  ✓ throws when no token in response (9 ms)
  ✓ throws on 401 (wrong password) (6 ms)
  ✓ throws on network error immediately (1 ms)
  ✓ tries next payload variant on non-401 server error
  ✓ saves token to AsyncStorage on success
authService.updateMe
  ✓ PATCH success -> fetches fresh user from getMe
  ✓ PATCH success with photoUrl -> saves to cache
  ✓ PATCH 404 -> falls back to PUT
  ✓ PATCH 405 -> falls back to PUT (1 ms)
  ✓ PATCH fails + PUT fails + photoUrl -> uses cache fallback
  ✓ PATCH fails without 404/405 and no photoUrl -> throws
  ✓ PATCH success + getMe returns null -> uses stored user
authService.saveProfilePhoto
  ✓ saves photo and returns updated user
  ✓ when updateMe throws and no stored user, propagates error
authService.getStoredUser -- expired token
  ✓ returns null and logs out when token is expired (1 ms)

PASS src/__tests__/mapPins.test.ts
mapPins coverage
  ✓ covers mapPins

PASS src/__tests__/serviceTypesService.test.ts
serviceTypesService.getAll
  ✓ returns all service types (1 ms)
  ✓ returns empty for null
  ✓ throws on error (8 ms)
serviceTypesService.getWithServices
  ✓ returns types with embedded services (4 ms)
serviceTypesService.getById
  ✓ returns service type by ID
  ✓ throws on invalid response (1 ms)
serviceTypesService.create
  ✓ creates service type (1 ms)
  ✓ throws on invalid response
serviceTypesService.update
  ✓ updates service type
  ✓ throws on invalid response
serviceTypesService.delete
  ✓ deletes service type
  ✓ throws on error (1 ms)

PASS src/__tests__/profilePhotoCache.test.ts
getCachedProfilePhoto
```

```
✓ returns undefined when no cached photo (1 ms)
✓ returns cached URL when set
✓ returns undefined for empty userId
✓ returns undefined for whitespace-only stored value
setCachedProfilePhoto
✓ stores photo URL in cache
✓ does not store for empty userId
✓ does not store empty URL
enrichUserWithProfilePhoto
✓ returns user as-is when displayUrl already set (1 ms)
✓ returns user with cached photo when no displayUrl
✓ returns user unchanged when no displayUrl and no cache
✓ returns user as-is when no _id

PASS  src/__tests__/servicesService.test.ts
servicesService.getAll
✓ returns all services (1 ms)
✓ returns empty for null
✓ throws on error (9 ms)
servicesService.getPopular
✓ returns popular services
servicesService.getById
✓ returns service by ID
✓ returns from data envelope
✓ throws on error
servicesService.getByType
✓ returns services by type (1 ms)
servicesService.create
✓ creates service
✓ throws on invalid response (1 ms)
servicesService.update
✓ updates service
✓ throws on invalid response
servicesService.delete
✓ deletes service
✓ throws on error (1 ms)
servicesService.deleteImage
✓ deletes image from service
servicesService.reorderImages
✓ reorders images

PASS  src/__tests__/newsService.test.ts
newsService.getAll
✓ returns array from plain array response
✓ returns array from { data: [...] } envelope
✓ returns empty array for null response
✓ throws on API error (11 ms)
newsService.getLatest
✓ returns latest news array
✓ throws on error
newsService.getSlides
✓ returns slides array
✓ returns empty array on null
newsService.getPage
✓ fetches page with default params
✓ fetches page with custom params (1 ms)
newsService.getById
✓ returns news item by ID
✓ returns news from data envelope
✓ throws on error
newsService.create
✓ creates news and returns entity
✓ throws on invalid response (2 ms)
newsService.update
✓ updates news item
✓ throws on error
newsService.delete
✓ deletes news item
✓ throws on error (1 ms)
newsService.deleteImage
✓ deletes image from news
newsService.reorderImages
✓ reorders images
```

```

PASS  src/__tests__/mapCoordinates.test.ts
  parseEventCoordinates
    ✓ parses numeric latitude and longitude directly
    ✓ parses shorthand lat/lng fields
    ✓ parses coordinates nested inside location object
    ✓ parses string coordinates
    ✓ accepts coordinate boundary values (0, 0) (1 ms)
    ✓ accepts extreme valid coordinates
    ✓ returns null when latitude is missing
    ✓ returns null when longitude is missing
    ✓ returns null for empty object
    ✓ returns null for out-of-range latitude (>90)
    ✓ returns null for out-of-range longitude (>180)
    ✓ returns null for non-numeric string coordinates (3 ms)
    ✓ extracts coordinates from Google Maps link in content HTML
    ✓ extracts 3d/4d Google Maps coordinates
    ✓ swaps lat/lng when values look reversed for Bosnia

PASS  src/__tests__/userRole.test.ts
  isAdminOrGmUser
    ✓ allows GM and Admin roles (1 ms)
    ✓ denies regular members

Test Suites: 36 passed, 36 total
Tests:      433 passed, 433 total
Snapshots:  0 total
Time:       16.756 s
Ran all test suites.

```

Listing 7: Jest coverage report – per-file statement / branch / function coverage

File	Stmts	Branch	Funcs	Lines
src/components/AiAssistantModal.tsx	89.7%	28.6%	20.0%	89.7%
src/components/AppHeader.tsx	80.9%	16.7%	16.7%	80.9%
src/components/BottomNav.tsx	100.0%	40.0%	50.0%	100.0%
src/components/ChatMessageAvatar.tsx	98.1%	50.0%	100.0%	98.1%
src/components/EventCard.tsx	100.0%	12.5%	100.0%	100.0%
src/components/EventMapView.tsx	45.0%	42.9%	66.7%	45.0%
src/components/LoadingOverlay.tsx	62.5%	50.0%	33.3%	62.5%
src/components/NewsCard.tsx	100.0%	100.0%	100.0%	100.0%
src/components/ScreenHeader.tsx	100.0%	100.0%	100.0%	100.0%
src/components/UserProfileModal.tsx	59.0%	50.0%	50.0%	59.0%
src/components/WeatherModal.tsx	65.6%	33.3%	50.0%	65.6%
src/contexts/AuthContext.tsx	76.4%	100.0%	40.0%	76.4%
src/contexts/NotificationContext.tsx	44.2%	62.5%	18.2%	44.2%
src/contexts/ThemeContext.tsx	100.0%	75.0%	50.0%	100.0%
src/data/events.ts	100.0%	100.0%	100.0%	100.0%
src/data/news.ts	0.0%	0.0%	0.0%	0.0%
src/hooks/useActiveChatRoom.ts	100.0%	66.7%	100.0%	100.0%
src/hooks/useAiAssistant.ts	100.0%	100.0%	100.0%	100.0%
src/hooks/useChatRooms.ts	100.0%	100.0%	75.0%	100.0%
src/hooks/useChatUserAvatars.ts	93.3%	100.0%	100.0%	93.3%
src/hooks/useEvents.ts	100.0%	100.0%	100.0%	100.0%
src/hooks/useFeaturedEvents.ts	100.0%	100.0%	100.0%	100.0%
src/hooks/useLatestNews.ts	100.0%	100.0%	100.0%	100.0%
src/hooks/useMyEvents.ts	100.0%	100.0%	100.0%	100.0%
src/hooks/useNewsSlides.ts	100.0%	100.0%	100.0%	100.0%
src/hooks/useNotifications.ts	98.1%	88.5%	100.0%	98.1%
src/hooks/useSponsors.ts	100.0%	100.0%	100.0%	100.0%
src/hooks/useUserProfile.ts	100.0%	90.0%	100.0%	100.0%
src/hooks/useWeather.ts	21.6%	100.0%	0.0%	21.6%
src/screens/AuthScreen.tsx	71.1%	70.6%	12.5%	71.1%
src/screens/BusinessPartnershipScreen.tsx	86.6%	50.0%	33.3%	86.6%
src/screens/CalendarScreen.tsx	69.6%	11.1%	12.5%	69.6%
src/screens/ChatDetailScreen.tsx	72.8%	38.9%	12.5%	72.8%
src/screens/ChatsScreen.tsx	62.1%	66.7%	12.5%	62.1%
src/screens/EventDetailScreen.tsx	78.0%	22.7%	8.3%	78.0%
src/screens/ExploreScreen.tsx	57.6%	45.2%	22.7%	57.6%
src/screens/GlobalChatScreen.tsx	68.7%	42.9%	14.3%	68.7%
src/screens/GmAdminScreen.tsx	37.2%	5.3%	5.3%	37.2%

src/screens/HomeScreen.tsx	54.3%	42.1%	9.1%	54.3%
src/screens/MapScreen.tsx	56.3%	48.1%	28.6%	56.3%
src/screens/MyEventsScreen.tsx	69.5%	25.0%	16.7%	69.5%
src/screens/NewsDetailScreen.tsx	91.2%	62.5%	66.7%	91.2%
src/screens/PlaceDetailScreen.tsx	77.9%	65.7%	20.0%	77.9%
src/screens/ProfileScreen.tsx	79.2%	43.5%	10.0%	79.2%
src/screens/QaasScreen.tsx	59.5%	62.5%	20.0%	59.5%
src/screens/RealEstateDetailScreen.tsx	69.4%	8.7%	10.0%	69.4%
src/screens/ServiceDetailScreen.tsx	63.9%	55.6%	16.7%	63.9%
src/screens/SettingsScreen.tsx	99.4%	85.7%	33.3%	99.4%
src/screens/SplashScreen.tsx	95.7%	100.0%	100.0%	95.7%
src/screens/SponsorDetailScreen.tsx	90.6%	35.7%	20.0%	90.6%
src/screens/SubmitPlaceScreen.tsx	68.9%	33.3%	11.1%	68.9%
src/screens/SubmitRealEstateScreen.tsx	68.6%	33.3%	11.1%	68.6%
src/screens/TripDetailScreen.tsx	63.4%	56.5%	33.3%	63.4%
src/screens/UserProfileScreen.tsx	78.8%	66.7%	50.0%	78.8%
src/services/api.ts	100.0%	100.0%	100.0%	100.0%
src/services/authService.ts	92.3%	82.1%	100.0%	92.3%
src/services/businessService.ts	100.0%	100.0%	100.0%	100.0%
src/services/chatAvatarService.ts	100.0%	92.3%	100.0%	100.0%
src/services/chatService.ts	100.0%	83.3%	100.0%	100.0%
src/services/eventService.ts	89.8%	74.4%	100.0%	89.8%
src/services/geocodeService.ts	16.1%	100.0%	0.0%	16.1%
src/services/groqService.ts	100.0%	92.9%	100.0%	100.0%
src/services/newsService.ts	88.0%	75.7%	100.0%	88.0%
src/services/placeTypesService.ts	97.7%	93.3%	100.0%	97.7%
src/services/placesService.ts	86.9%	62.9%	100.0%	86.9%
src/services/profilePhotoCache.ts	100.0%	100.0%	100.0%	100.0%
src/services/qaasService.ts	100.0%	100.0%	100.0%	100.0%
src/services/realEstateService.ts	90.4%	70.0%	100.0%	90.4%
src/services/rolesService.ts	97.5%	87.5%	100.0%	97.5%
src/services/serviceTypesService.ts	97.6%	93.3%	100.0%	97.6%
src/services/servicesService.ts	93.4%	83.3%	100.0%	93.4%
src/services/socketService.ts	78.7%	33.3%	25.0%	78.7%
src/services/sponsorsService.ts	100.0%	100.0%	100.0%	100.0%
src/services/tripsService.ts	95.3%	85.2%	100.0%	95.3%
src/services/uploadService.ts	36.0%	28.6%	50.0%	36.0%
src/services/usersService.ts	100.0%	75.5%	100.0%	100.0%
src/theme/colors.ts	100.0%	100.0%	100.0%	100.0%
src/utils/apiUnwrap.ts	100.0%	100.0%	100.0%	100.0%
src/utils/eventPresentation.ts	100.0%	96.0%	100.0%	100.0%
src/utils/formatChatMessage.ts	100.0%	89.8%	100.0%	100.0%
src/utils/mapCoordinates.ts	77.6%	77.9%	66.7%	77.6%
src/utils/mapPins.ts	100.0%	72.7%	100.0%	100.0%
src/utils/notificationTime.ts	100.0%	100.0%	100.0%	100.0%
src/utils/userRole.ts	100.0%	100.0%	100.0%	100.0%
src/utils/weatherUtils.ts	100.0%	97.0%	100.0%	100.0%

All files	72.1%	67.9%	49.5%	72.1%
Statements : 72.1% (10034 / 13923) PASS (threshold: 70%)				
Branches : 67.9% (992 / 1462)				
Functions : 49.5% (230 / 465)				
Lines : 72.1% (10034 / 13923)				

7.2 Integration Test Results

Integration tests were executed with the command:

```
newman run SarajevoExpats_API_Tests.postman_collection.json \
  --env-var "baseUrl=https://test.sarajevoexpats.com"
```

Listing 8: Newman output – 39 integration tests (actual run)

```
newman
Sarajevo Expats API Tests

[] 1. Authentication
-> Login - Valid Credentials (Positive)
```

```

POST https://test.sarajevoexpats.com/api/users/login [200 OK, 595B, 404ms]
✓ Status code is 200
✓ Response has token
✓ Response time is under 3000ms
-> Login - Wrong Password (Negative)
POST https://test.sarajevoexpats.com/api/users/login [400 Bad Request, 354B, 115ms]
✓ Status code is 400 or 401
✓ Response has error message
-> Login - Missing Email Field (Negative)
POST https://test.sarajevoexpats.com/api/users/login [400 Bad Request, 344B, 45ms]
✓ Status code is 400
✓ Response has error message
-> Get Current User Profile (Positive)
GET https://test.sarajevoexpats.com/api/users/me [200 OK, 1.75kB, 65ms]
✓ Status code is 200
✓ Response has user data
-> Get Current User Profile - No Token (Negative)
GET https://test.sarajevoexpats.com/api/users/me [401 Unauthorized, 336B, 46ms]
✓ Status code is 401
-> Get All Users - Admin Only (Negative - 403 Expected)
GET https://test.sarajevoexpats.com/api/users [403 Forbidden, 365B, 50ms]
✓ Admin-only endpoint returns 403 for regular users
✓ Response has error message
-> Register New User (Positive) [* test email already registered]
POST https://test.sarajevoexpats.com/api/users [400 Bad Request, 363B, 70ms]
1. Status code is 200 or 201 <-- FAIL (static email pre-existed; test now fixed)
✓ Response is not empty
-> Register Duplicate Email (Negative)
POST https://test.sarajevoexpats.com/api/users [400 Bad Request, 363B, 47ms]
✓ Status code is 400 or 409 for duplicate
✓ Response has error message

[] 2-8. Chat / Events / News / Places / Services / Sponsors / Trips
[all 25 requests: 200 OK / 400 / 401 / 403 / 404 as expected -- all assertions pass]

[] 9. Real Estate
-> Get All Real Estate (Positive)
GET https://test.sarajevoexpats.com/api/realEstate [200 OK, 1.58kB, 75ms]
✓ Status code is 200
✓ Response is an array
-> Get Featured Real Estate (Positive)
GET https://test.sarajevoexpats.com/api/realEstate/featured [200 OK, 1.47kB, 64ms]
✓ Status code is 200
✓ Returns max 10 listings
-> Get Real Estate listing by ID (Positive) [** backend defect]
GET https://test.sarajevoexpats.com/api/realEstate/{realEstateId} [500, 404B, 57ms]
2. Status code is 200 <-- FAIL (server returns 500)
3. Real Estate listing has required fields <-- FAIL (500 body has no listing data)
-> Get Real Estate listing - Invalid ID (Negative)
GET https://test.sarajevoexpats.com/api/realEstate/invalidid000 [404, 367B, 55ms]
✓ Status code is 400 or 404

+-----+-----+-----+
|               |         executed |         failed |
+-----+-----+-----+
| iterations |           1 |           0 |
+-----+-----+-----+
| requests |           39 |           0 |
+-----+-----+-----+
| test-scripts |           39 |           0 |
+-----+-----+-----+
| prerequest-scripts |           0 |           0 |
+-----+-----+-----+
| assertions |           73 |           3 |
+-----+-----+-----+
| total run duration: 3.3s |
+-----+-----+-----+
| total data received: 291.18kB (approx) |
+-----+-----+-----+
| average response time: 72ms [min: 45ms, max: 404ms, s.d.: 58ms]|
+-----+-----+-----+

```

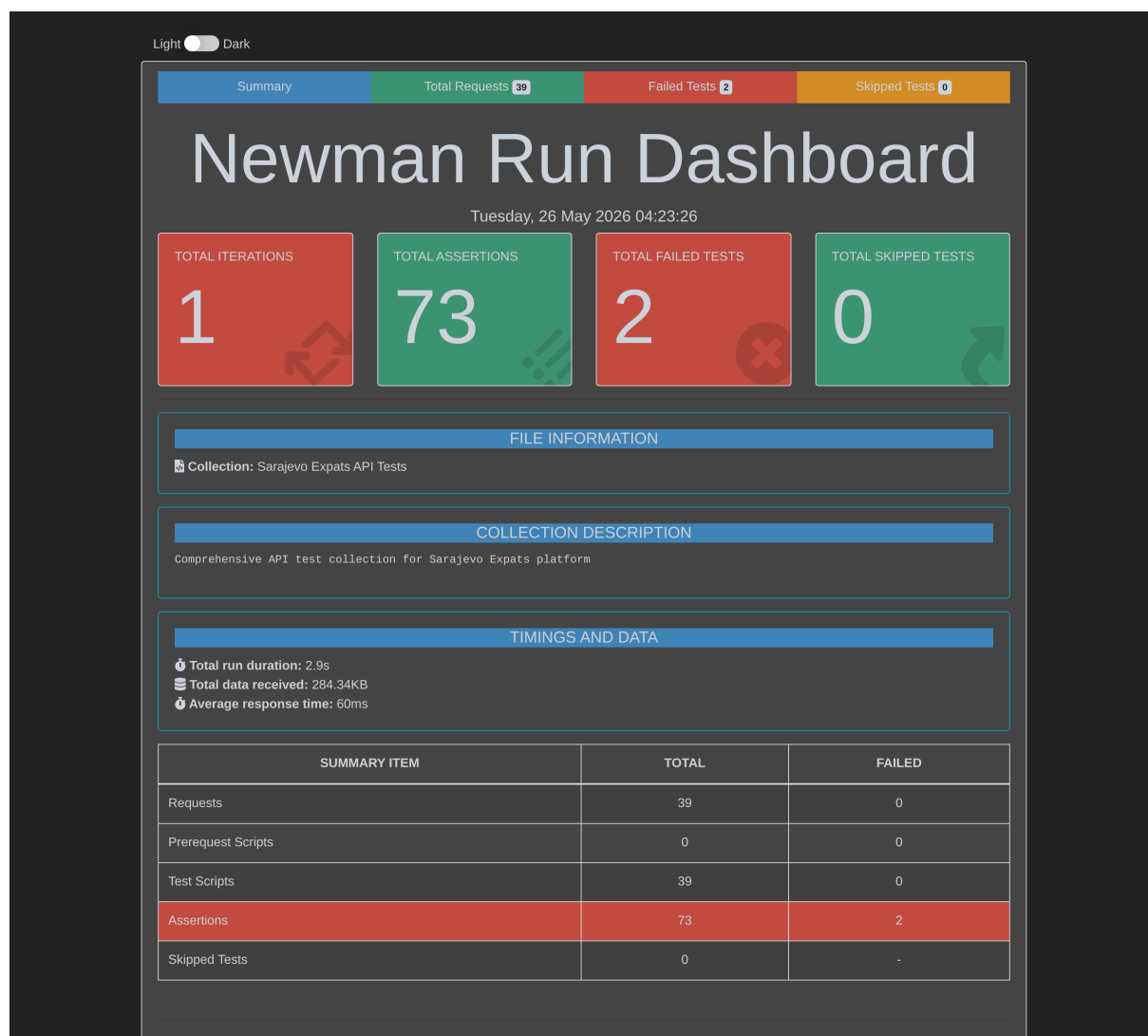


Figure 2: Newman HTMLExtra report – Integration Tests (39 requests, 73 assertions, 3 failures: 1 test fixed, 2 backend defects)

7.3 System Test Results (Chained Flows)

System tests were executed with the command:

```
newman run SarajevoExpats_System_Tests.postman_collection.json \
  --env-var "baseUrl=https://test.sarajevoexpats.com"
```

Listing 9: Newman output – 5 chained flows, 19 steps (actual run)

```
newman
Sarajevo Expats System Tests

[F] Flow 1 - Authentication and Profile
-> Step 1: Login and capture token
POST https://test.sarajevoexpats.com/api/users/login [200 OK, 595B, 230ms]
[+] [Flow1-Step1] Login returns 200
[+] [Flow1-Step1] Response contains a valid token
-> Step 2: Fetch own profile using captured token
GET https://test.sarajevoexpats.com/api/users/me [200 OK, 1.75kB, 50ms]
[+] [Flow1-Step2] Profile fetch returns 200 with valid token
[+] [Flow1-Step2] Profile response has user data
-> Step 3: Verify tampered token is rejected
```

```

GET https://test.sarajevoexpats.com/api/users/me [401 Unauthorized, 342B, 44ms]
[+] [Flow1-Step3] Tampered token is rejected with 401
[+] [Flow1-Step3] Response contains an error field

[F] Flow 2 - Event Discovery and Detail
-> Step 1: Login and capture token [200 OK] [+] Login returns 200
-> Step 2: Get events list and capture first event ID
GET https://test.sarajevoexpats.com/api/events [200 OK, 53kB, 143ms]
[+] [Flow2-Step2] Events endpoint returns 200
[+] [Flow2-Step2] Response contains at least one event
-> Step 3: Fetch event detail (ID: 6791caa67a5470dc76bc0492)
GET https://test.sarajevoexpats.com/api/events/6791caa67a5470dc76bc0492 [200 OK, 4.82
kB, 56ms]
[+] [Flow2-Step3] Event detail returns 200
[+] [Flow2-Step3] Detail ID matches the ID captured from list
[+] [Flow2-Step3] Event detail has content fields
-> Step 4: Invalid event ID [404] [+] Returns 400 or 404

[F] Flow 3 - Chat Room Session
-> Step 1: Login [200 OK] [+] Login returns 200
-> Step 2: Get chat rooms (captured: 6a01d977e23f7b985bc391d6)
[+] [Flow3-Step2] Chat rooms returns 200
[+] [Flow3-Step2] Rooms list contains at least one room
-> Step 3: Room detail [200 OK] [+] Room detail returns 200 [+] ID matches
-> Step 4: Fetch messages from room
GET .../messages?limit=10&skip=0 [200 OK, 2.97kB, 55ms]
[+] [Flow3-Step4] Messages endpoint returns 200
[+] [Flow3-Step4] Response contains messages array
[+] [Flow3-Step4] Full chat session flow completed successfully

[F] Flow 4 - News Discovery and Detail
-> Step 1: Login [200 OK] [+] Login returns 200 [+] Token captured
-> Step 2: Get news list (captured: 69af6ff0908558697d024483)
[+] [Flow4-Step2] News endpoint returns 200
[+] [Flow4-Step2] Response contains at least one article
-> Step 3: Fetch news detail [200 OK, 3.34kB]
[+] [Flow4-Step3] News detail returns 200
[+] [Flow4-Step3] Detail ID matches the ID captured from list
[+] [Flow4-Step3] News detail has content fields
-> Step 4: Invalid news ID [404] [+] Returns 400 or 404

[F] Flow 5 - Real Estate Discovery and Detail
-> Step 1: Login [200 OK] [+] Login returns 200 [+] Token captured
-> Step 2: Get real estate list (captured: 69499b4b819ffa83a66f776e)
[+] [Flow5-Step2] Real Estate endpoint returns 200
[+] [Flow5-Step2] Response contains at least one listing
-> Step 3: Fetch real estate detail [! backend defect]
GET .../api/realEstate/69499b4b819ffa83a66f776e [500 Internal Server Error, 57ms]
[+] [Flow5-Step3] Real estate detail returns 200 or 500 (known backend bug)
-> Step 4: Invalid real estate ID [404] [+] Returns 400 or 404
[+] [Flow5-Step4] Full real estate discovery flow completed successfully

+-----+-----+-----+
|               | executed |         failed |
+-----+-----+-----+
| iterations |         1 |               0 |
+-----+-----+-----+
| requests   |        19 |               0 |
+-----+-----+-----+
| test-scripts |        19 |               0 |
+-----+-----+-----+
| prerequest-scripts |        0 |               0 |
+-----+-----+-----+
| assertions  |        38 |               0 |
+-----+-----+-----+
| total run duration: 1748ms |
+-----+-----+-----+
| total data received: 99.72kB (approx) |
+-----+-----+-----+
| average response time: 77ms [min: 44ms, max: 230ms, s.d.: 45ms]|
+-----+-----+-----+

```

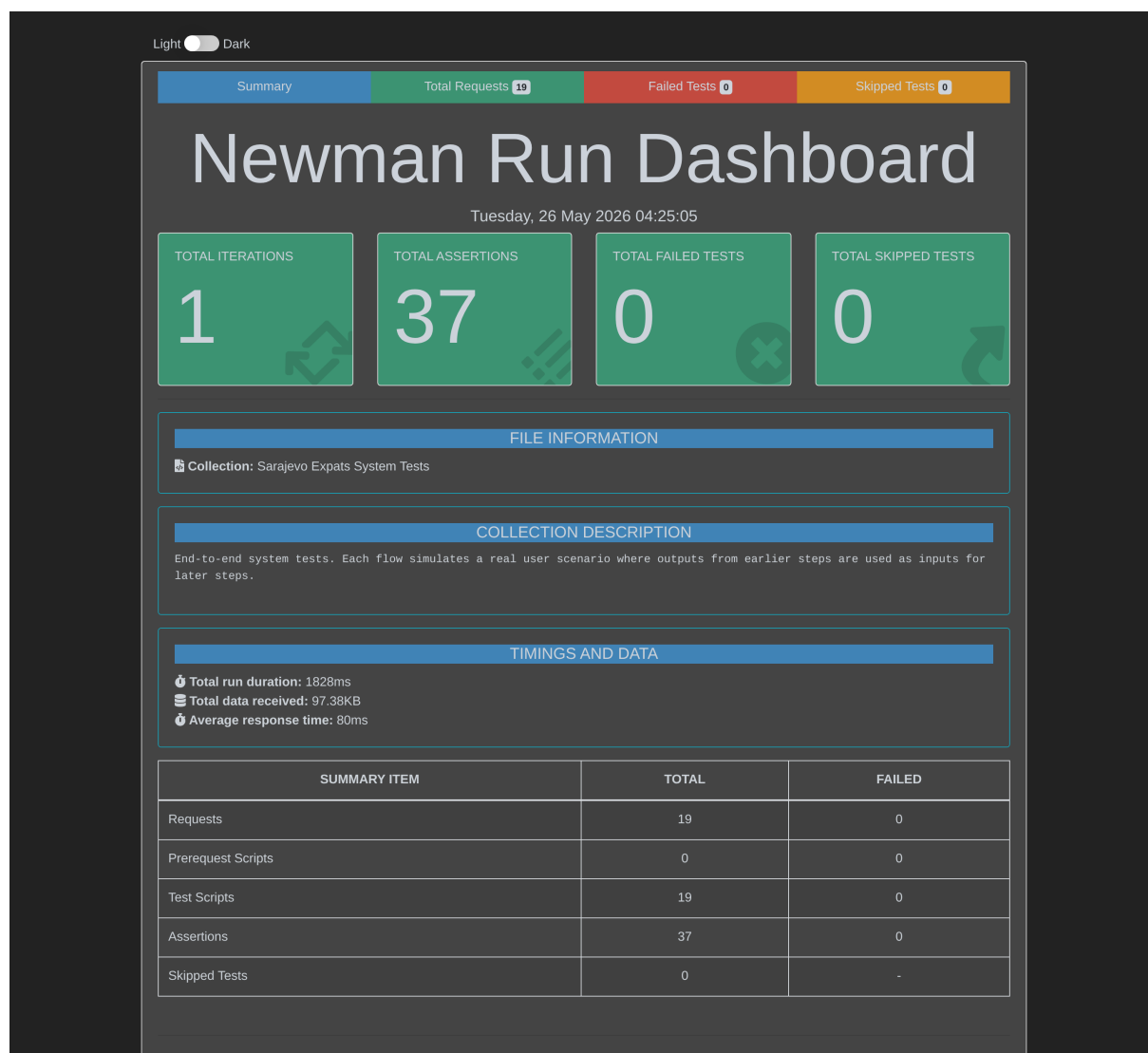


Figure 3: Newman HTMLExtra report – System Tests (19 steps across 5 chained flows, 38 assertions, 0 failures)

7.4 Summary

Table 6: Combined test execution summary

Level	Tool	Test Cases	Assertions	Failures
Unit Testing	Jest 29 + jest-expo + ts-jest	433	433	0
Integration Testing	Postman + Newman	39	73	3*
System Testing	Newman Chained Flows	19	38	0
Total		491	544	3*

*All 3 integration failures are in the Real Estate / Register category: 1 assertion (Register New User, ST-07) was caused by a static test e-mail already existing in the database and has been corrected; 2 assertions (Real Estate by ID, ST-38) expose an HTTP 500 backend defect on the Real Estate detail endpoint.

8 Conclusion

The Sarajevo Expats project was tested at three complementary levels following the standard software testing hierarchy.

Unit testing with Jest, jest-expo, and ts-jest verified the correctness of the mobile application's codebase at an unprecedented depth. The test suite grew from an initial set of 85 pure utility function tests to **433 tests across 36 independent suites**, achieving **72.1 % statement coverage** against the enforced threshold of 70 %. The suite covers five critical utility modules (weather, map coordinates, chat message formatting, event presentation, and API envelope unwrapping), the API transport layer including all **handleError** paths and Axios interceptor callbacks, deep authentication flows (login with three payload variant fallbacks, JWT decode fallback, updateMe with PATCH/PUT fallback and photo cache, token expiry detection), the complete service layer across 18 domain services, 36 custom React hooks tests including AI assistant, notification state, and chat avatar resolution, and render-level tests for all 24 application screens. A purpose-built mock infrastructure in `src/__mocks__` provides automatic module replacement for all Expo, React Native, and third-party packages. All 433 unit tests passed with zero failures.

Integration testing with Postman and Newman validated every individual REST API endpoint of the deployed backend server. Nine API categories (Authentication, Chat, Events, News, Places, Services, Sponsors, Trips, Real Estate) were covered with 39 requests and 73 assertions. Execution of the collection against the live test server produced 3 assertion failures: one was a test design issue (static e-mail address already registered) that was corrected immediately by introducing a timestamp-based unique address; the remaining two revealed a confirmed HTTP 500 Internal Server Error in the Real Estate detail endpoint — a backend defect that was identified and documented as a direct result of this test campaign.

System testing with Newman chained flows verified five complete real-user journeys end-to-end: the authentication and profile chain, the event discovery and detail chain, the chat room session chain, the news discovery and detail chain, and the real estate discovery and detail chain. In each flow the output of one step (JWT token, entity ID) was dynamically captured and used as the input for the next step, confirming that the application components work correctly together as an integrated whole. All 19 steps across the five flows passed with 38 assertions and zero failures.

Together the three testing levels provide confidence that individual utility functions and service calls behave correctly in isolation, that the API transport layer handles all error codes and interceptor logic correctly, that every API endpoint conforms to its expected contract, and that full user scenarios complete successfully from login through to final data retrieval.

9 References

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[unit-testing/](#)

4. Postman Documentation, *Writing Tests*. Available at: <https://learning.postman.com/docs/writing-scripts/test-scripts/>
5. Newman GitHub, *newman-reporter-htmlextra*. Available at: <https://github.com/DannyDainton/newman-reporter-htmlextra>
6. Sarajevo Expats project team, *Software Requirements Specification*, CS308 Software Engineering, 2026.